Interview question and respose

- Question1: How do you envision the audit log system improving the overall integrity of online examinations?
- Solve: I envision it by providing an immutable, detailed, and chronological record of every user action, which helps instructors easily spot suspicious behaviors and deters students from attempting academic dishonesty.

- Question2: What kind of challenges do you expect when storing and managing large volumes of real-time event logs during peak exam periods?
- solve: The main challenges include database performance degradation, high network traffic, and ensuring storage scalability, which we can solve by implementing efficient indexing and log archiving strategies.

- Question3: Does the system automatically flag a student's exam session when they switch browser tabs multiple times?
- solve: Yes, the system detects tab-switching events via JavaScript listeners and immediately records a violation flag in the audit log database.

- Question4: Is the audit log data encrypted at rest to prevent unauthorized tampering or modifications?
- solve: Yes, all audit log records are encrypted using standard cryptographic algorithms in the database to guarantee data integrity.

- Question5: Should the system block the student's exam automatically upon detecting a violation, or should it only notify the proctor in real time?

- solve: The system should notify the proctor in real time and let them decide whether to block the student or allow them to continue, preventing false-positive automatic lockouts.

- Question6: Do you prefer storing the generated audit logs in a relational database like MySQL or a specialized log management system?
- solve: We prefer using a relational database for structured student-exam mapping, combined with structured log tables optimized for quick querying and filtering by the proctor.

- Question7: You mentioned that the system tracks tab-switching; exactly how does it capture this event if the browser is minimized?
- solve: It utilizes the HTML5 Page Visibility API along with window blur/focus event listeners, which trigger an immediate AJAX request to log the exact timestamp when the user leaves the exam page.

- Question8: Can you elaborate on how administrators can export these audit logs for institutional compliance or legal reviews?
- solve: Administrators can access a dedicated dashboard report section where they can filter logs by student ID or exam session and export the complete audit trail into CSV or PDF formats.

- Question9: How will the user interface (dashboard) present complex log data so that human proctors aren't overwhelmed during a live exam?
- solve: The dashboard will use a color-coded severity system (e.g., green for normal, yellow for warnings, red for high-risk violations) alongside summary widgets and a live event stream.

- Question10: If a network interruption occurs on the client side, how does the system prevent the loss of critical audit log events?
- solve: The system caches unsent audit logs in the browser's local storage temporarily and automatically syncs them back to the server once the network connection is restored.